Md. Atik Mouhtasim

Aspiring AI/ML Researcher | Web & Mobile App Developer

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Summary

Computer Science student specializing in AI systems, dataset quality, and cross-platform application development. Hands-on experience validating autonomous-driving datasets for self-supervised monocular TTC estimation and building web and mobile applications using Flutter, React Native, Next.js, Firebase, and Supabase.

Education

B.Sc. **Rajshahi University of Engineering & Technology**, B.Sc. in Computer Science and Engineering (CSE) – Rajshahi — Dec 2021 – Aug 2026
CGPA: 3.53/4.0

HSC **Notre Dame College**, HSC in Science – Dhaka — Jan 2018 – Jan 2020
GPA: 5.0/5.0

SSC **Thakurgaon Government Boys High School**, SSC – Thakurgaon — Jan 2010 – Jan 2018
GPA: 5.0/5.0

Experience

RUET Computing Society, General Secretary — Nov 2025 – Sept 2026

- Led execution of plans by distributing responsibilities and managing manpower across multiple activities.
- Handled internal coordination and issue resolution to maintain a productive club environment.
- Oversaw timelines to ensure all administrative and technical activities are completed on time.

Texlab IT, Industrial Trainee — Mar 2025 – Apr 2025

- Gained hands-on experience in a professional software development environment
- Learned collaboration, issue tracking, and async communication using GitHub and Slack
- Developed a strong foundation in professionalism, workplace ethics and industry standard workflows

Projects

MediTrack — Aug 2026 – Sept 2026

Tech Stack: Flutter, Dart, Firebase, ML Kit, Gemini, OpenRouter

- Built a bilingual medicine-management app for prescription scanning, dosage scheduling, adherence tracking, stock and refill alerts, and family profiles.
- Designed a two-stage prescription pipeline combining on-device ML Kit OCR with Gemini extraction, schema validation, and mandatory user review before medication data is saved.
- Implemented Firebase authentication and persistence, local reminders, an offline SQLite medicine catalog, and **a context-aware AI assistant**.

WasteGuard — May 2025 – present

Tech Stack: React Native (Expo), TypeScript, Supabase, NativeWind

- Built a cross-platform food-waste management app for pantry inventory, expiry tracking, waste analytics, and recipe suggestions based on soon-to-expire ingredients.
- Implemented cloud synchronization, push notifications, Supabase authentication and data storage, and TanStack Query server-state management.

Asteroid Odyssey — Sept 2025 – Oct 2025

Tech Stack: NASA APIs, Next.js, TypeScript, Three.js, D3.js

- Led a five-person team as team leader and lead developer for NASA Space Apps 2025.
- Built a 3D asteroid-impact simulator that explores more than 100 objects and models mitigation strategies and global effects over a 50-year timeline.
- Integrated NASA NEO, JPL Small-Body Database, Sentry, and Fireball data with interactive visualizations for orbital exploration, threat assessment, and impact analysis.

Prayer.time Feb 2026

Tech Stack: React, TypeScript, Vite, Tailwind CSS

- Built a privacy-first browser extension for prayer-time and Ramadan countdowns without continuous GPS tracking.
- Integrated prayer-time calculation through adhan, OpenStreetMap Nominatim city search, 11 calculation methods, and per-prayer time calibration.
- Implemented service-worker alarms, cross-tab browser notifications, and in-app alerts; published releases for Edge, Firefox, and Chromium browsers.

UniCompanion 2024

Tech Stack: React Native (Expo), Supabase, TypeScript, NativeWind

- Built a cross-platform academic management application for students and teachers using React Native, Expo, and Supabase.
- Implemented attendance tracking with CSV and Excel export, class and content management, CT and lab assessments, course enrollment, and in-app messaging.
- Integrated Google OAuth, cloud storage, and NativeWind interface components to support authenticated teacher and student workflows.

Research

MonoTTC: Object-Level Time-to-Contact Estimation under Limited Supervision Aug 2026

B.Sc. thesis supervised by Farjana Parvin, Department of CSE, RUET

- Developed a two-stage PyTorch pipeline combining self-supervised ResNet-18 and PoseNet geometry pretraining on temporal nuScenes triplets with ROI-supervised TTC adaptation using frozen RAFT-Small.
- Derived audited metric TTC targets from nuScenes 3D tracks, calibrated ego poses, and camera geometry using scene-disjoint splits, a 25% label budget, and a single 16 GB GPU.
- Across five seeds and 87 validation scenes, geometric consistency produced a 5.33% descriptive reduction in scene-balanced relative TTC error from 0.4131 ± 0.0186 to 0.3911 ± 0.0178.
- Reduced absolute MAE for urgent 0–2 s TTC targets from 2.1075 s to 1.6826 s; the paired 95% confidence interval included zero, so the overall improvement was not statistically significant.
- Achieved AUROC of 0.8667 ± 0.0256 and AUPRC of 0.2080 ± 0.0493 for oracle risk classification of critical targets with TTC ≤ 2.0 s.

Skills

Programming Languages: Python, JavaScript, TypeScript, Dart, C++, HTML, CSS, PHP

AI/ML: PyTorch, torchvision, Kornia, NumPy, SciPy, Scikit-learn, OpenCV, Seaborn

Frontend: React, Next.js, React Native (Expo), Flutter, Three.js, D3.js

Backend: Supabase/PostgreSQL, Firebase, SQLite, MySQL, Laravel

Tools: Linux, Docker, Git, GitHub, Tectonic

Agents: OpenCode TUI, Codex-CLI, Claude Code, Deepseek-harness, OpenChamber

Certifications

- Industrial Trainee, Texlab IT — Certificate
- EDGE Course (PHP, Laravel), Bangladesh Computer Council and RUET — Certificate
- Supervised Machine Learning: Regression and Classification, Coursera — Certificate
- Web Development Bootcamp, Programming Hero — Certificate

References

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